Cell Biology & Biochemistry

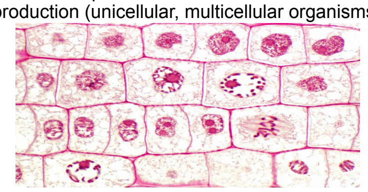
Topics Covered
Biochemistry
Biomolecules and Molecular Organization
Biological Macromolecules:
- Carbohydrates
- Lipids
- Proteins
- Nucleic acids and Nucleotides
- Identification of Biomolecules (Practical)

Cell Biology
Cell ultra-structure
Biological membranes: Structure and functions
Cell Division
DNA Replication
Protein Biosynthesis
Cellular communication
Bioenergetics
- Enzymes: General properties and mechanism of action
- Cellular respiration

Cell Biology

Cell Division

- "Cells arise only by the division of a previously existing cell"
- The process in which a parent cell divides in to two or more cells
- Cell division is required for
  - growth and replacement
  - reproduction (unicellular, multicellular organisms)



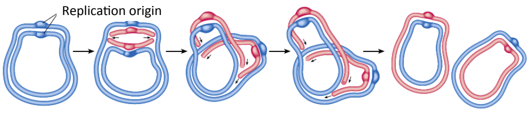
Cell Biology

Cell Division – Topics Covered

- Cell Division in Prokaryotes
- Cell Division in Eukaryotes
  - Cell Cycle
  - Mitosis
  - Meiosis

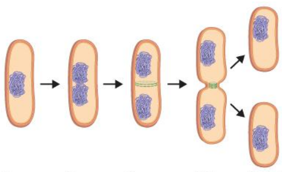
Cell Biology

Cell Division: Cell division in Prokaryotes
- Cell divides in to two nearly equal halves - Binary fission
- Growth of the prokaryote cell twice its' size induces cell division
- Genome replicates early before fission
- Replication of genome/DNA starts at the **replication origin**
- Nearly 22 enzymes and proteins are involved
- After replication, two copies of the DNA are attached side by side in the plasma membrane



Cell Biology

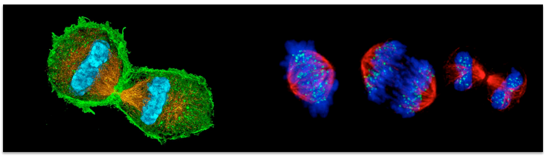
Cell Division: Cell division in Prokaryotes
- Fission starts between the points where the two genomes are attached to the plasma membrane
- Initially the membrane pinches inward in the middle of the cell
- A new cell wall and a plasma membrane grows between the two genomes



Cell Biology

Cell Division: Cell division in Eukaryotes

- Two types of cell division
  - mitosis: produces somatic cells
  - meiosis: produces gametes (in reproduction)
- Mitosis occurs as a part of the **cell cycle**

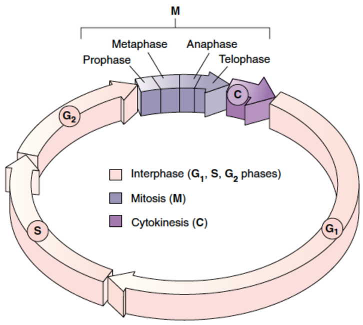


Cell Biology

Cell Division: Cell division in Eukaryotes

Cell Cycle

- Series of events that take place in a cell leading to **duplication of its DNA** and **division** to produce **two daughter cells**
- Consists of five phases (**G_1, S, G_2, M** and **C**)



Cell Biology

Cell Division: Cell division in Eukaryotes

Cell Cycle

- Gap 0 (G_0) - resting phase, cell has left the cycle and has stopped dividing
- Gap 1 (G_1) - primary growth phase of the cell (increase the size), major portion of the cell's life span
- Synthesis (S)- genome/DNA replication
- Gap 2 (G_2) - second growth phase, preparations are made for genomic separation, mitochondria and other organelles replicate and microtubules begin to assemble at a spindle

Cell Biology

Cell Division: Cell division in Eukaryotes

Cell Cycle

- G_1, S, and G_2 together constitute **interphase,** the portion of the cell cycle between cell divisions
- Mitosis (M) – the two daughter genomes are separated (microtubular apparatus binds to the chromosomes, and moves the sister chromatids apart)
- Cytokinesis (C) - cytoplasm divides, creating two daughter cells
- Duration of the cycle can vary from a few minutes (fruit fly embryo cells) to a year (Human liver cells)

Cell Biology

Cell Division: Cell division in Eukaryotes

Cell Cycle: Regulation of the cell cycle

- What happens if nuclear DNA is not replicated before the nucleus begins to divide?
- What should the cell do if the DNA replication is slowed?
- What should the cell do if the DNA is damaged?
- The cell employs a series of molecular breaks in a feedback mechanism to prevent any of the above - **Checkpoints**

Cell Biology

Cell Division: Cell division in Eukaryotes

Cell Cycle: Regulation of the cell cycle

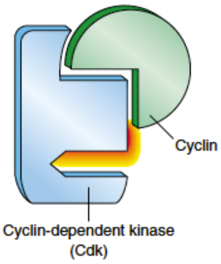
- **Checkpoints** ensure proper division of the cell
- Three major checkpoints in the cell cycle
  - G_1 checkpoint
  - G_2 checkpoint
  - Mitosis/Metaphase checkpoint
- Checkpoints are controlled/regulated by a complex of enzymes – Cyclin dependent kinases (Cdk)

Cell Biology

Cell Division: Cell division in Eukaryotes

Cell Cycle: Regulation of the cell cycle

- Activity of Cdk depends on **Cyclin** proteins
- Binding of Cyclins to Cdks, enables Cdks to function as enzymes
- Cdks bound to Cyclin trigger the the steps of the cell cycle to proceed from one to another

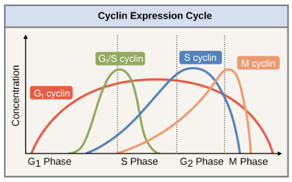


Cell Biology

Cell Division: Cell division in Eukaryotes

Cell Cycle: Regulation of the cell cycle

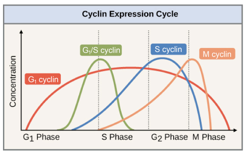
- Different types of cyclins regulate the different checkpoints
- Concentration of different cyclins change cyclically
- Cyclins are destroyed and resynthesized during each turn of the cell cycle

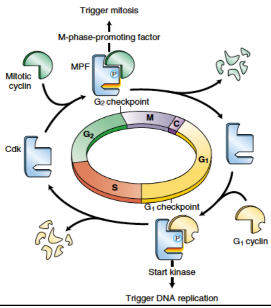


Cell Biology

Cell Division: Cell division in Eukaryotes

Cell Cycle: Regulation of the cell cycle





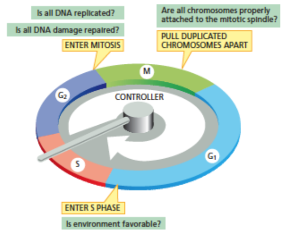
Cell Biology

Cell Division: Cell division in Eukaryotes

Cell Cycle: Regulation of the cell cycle

G_1 checkpoint

- Happens before the beginning of S phase
- Checks for the suitable cellular environmental conditions for replication e.g. condition of DNA
- If conditions are not favorable, the cell may enter the resting state (G_0)

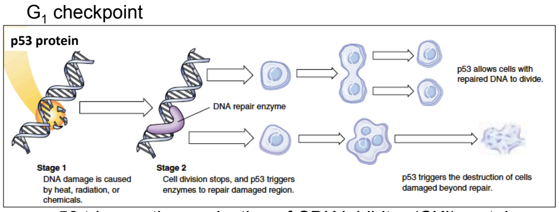


Cell Biology

Cell Division: Cell division in Eukaryotes

Cell Cycle: Regulation of the cell cycle

G_1 checkpoint



- p53 triggers the production of CDK inhibitor (CKI) protein
- CKI proteins bind to Cdk-cyclin complexes and block the activity

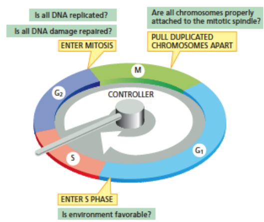
Cell Biology

Cell Division: Cell division in Eukaryotes

Cell Cycle: Regulation of the cell cycle

G_2 checkpoint

- Takes place after the S phase
- Check the cell's DNA, making sure it is structurally intact and properly replicated
- May pause at this point to allow time for DNA repair, if necessary
- Prevents the cell from entering the mitosis (M) phase with genomic DNA damage



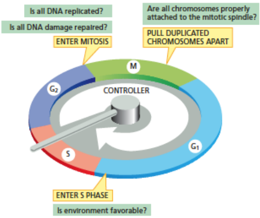
Cell Biology

Cell Division: Cell division in Eukaryotes

Cell Cycle: Regulation of the cell cycle

Mitosis checkpoint

- Takes place in the middle of mitosis (metaphase)
- Ensures that all of the
  - chromosomes are properly aligned
  - and the kinetochores are correctly attached to spindle



Cell Biology

Cell Division: Cell division in Eukaryotes

Two distinct types of cell division

- **mitosis**
  - a vegetative division
  - each daughter cell is genetically identical to the parent cell
- **meiosis**
  - reductive cell division (number of chromosomes in the daughter cells are reduced by half)
  - results in four **haploid (n)** daughter cells
  - produce haploid gametes
  - undergoes one round of DNA replication, two divisions

Cell Biology

Cell Division: Cell division in Eukaryotes

Mitosis

Five stages - Prophase
- Prometaphase
- Metaphase
- Anaphase
- Telophase

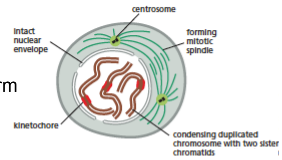
- Cytokinesis (part of the cell cycle)

Cell Biology

Cell Division: Cell division in Eukaryotes

Mitosis – Prophase

- Nucleolus disappears
- Chromosomes condense
- Mitotic spindle begins to form between centrosomes
- Two centrosomes move towards the opposite sides

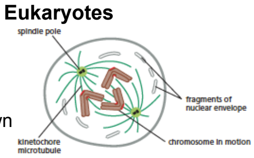


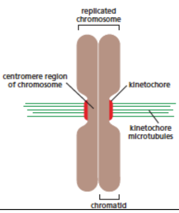
Cell Biology

Cell Division: Cell division in Eukaryotes

Mitosis – Prometaphase

- Nuclear membranes break down
- Mitotic spindle attach to the kinetochores of chromosomes
- Kinetochore - a complex of proteins positioned at the centromere



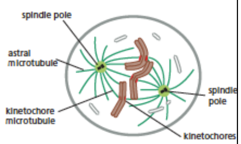


Cell Biology

Cell Division: Cell division in Eukaryotes

Mitosis – Metaphase

- Chromosomes are aligned along the metaphase plate at the equator of the cell
- Each chromosome has two microtubules extending from its kinetochore
- Three types of microtubules are present:
  - Kinetochore microtubules attach the chromosomes to the spindle pole
  - interpolar microtubules extend from the spindle pole across the equator
  - astral microtubules extend from the spindle pole to the cell membrane

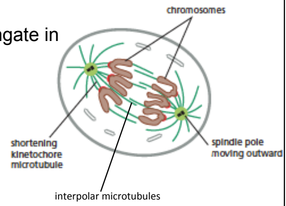


Cell Biology

Cell Division: Cell division in Eukaryotes

Mitosis – Anaphase

- Kinetochore microtubules shorten pulling chromosomes towards opposite poles
- Interpolar microtubules elongate in preparation of cytokinesis

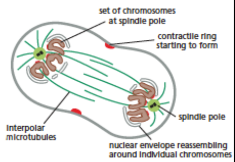


Cell Biology

Cell Division: Cell division in Eukaryotes

Mitosis – Telophase

- Chromosomes reach poles of cell
- Kinetochores begin to disappear
- Interpolar microtubules continue to elongate in preparation of cytokinesis
- Nuclear membranes reform
- Nucleolus reappears
- Chromosomes disappear

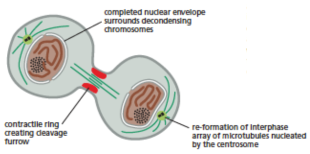


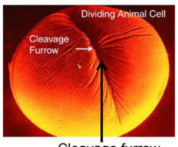
Cell Biology

Cell Division: Cell division in Eukaryotes

Cytokinesis

- Cell divides creating two daughter cells
- Animal cells – cleavage furrow forms at the equator of the cell, pinches inward until cells divide in to two, cleavage furrow produced by constricting actin and myosin filaments (Contractile ring)





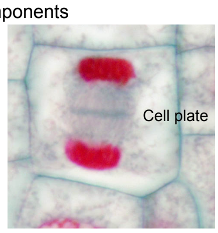
Cleavage furrow

Cell Biology

Cell Division: Cell division in Eukaryotes

Cytokinesis

- Plant cells – cell plate forms dividing the the cells, cell plate is laid by vesicles (from Golgi apparatus) containing membrane components



Cell Biology

Cell Division: Cell division in Eukaryotes

Meiosis

- The cell undergoes two divisions
- Two major stages - meiosis I
  - meiosis II
- Meiosis I - segregates homologous chromosomes producing two haploid cells
- Meiosis II – produces four haploid cells from the two haploid cells

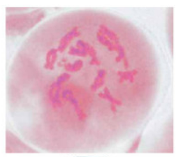
Cell Biology

Cell Division: Cell division in Eukaryotes

Meiosis: Meiosis I

Prophase I

- The homologous chromosomes pair and exchange DNA to form recombinant chromosomes
- Prophase I is divided into five phases:

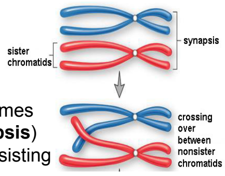


Cell Biology

Cell Division: Cell division in Eukaryotes

Meiosis: Meiosis I

Prophase I



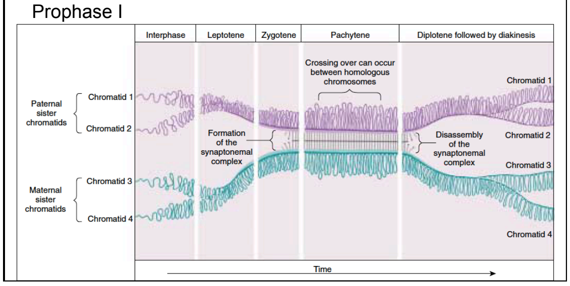
- **Leptotene**: chromosomes start to condense
- **Zygotene**: homologous chromosomes become closely associated (**synapsis**) to form pairs of chromosomes consisting of four chromatids (assisted by synaptonemal complex)
- **Pachytene**: formation of **chiasmata (attachment) between** pairs of homologous chromosomes

Cell Biology

Cell Division: Cell division in Eukaryotes

Meiosis: Meiosis I

Prophase I



Cell Biology

Cell Division: Cell division in Eukaryotes

Meiosis: Meiosis I

Prophase I

- Diplotene: synaptonemal complex degrades, homologous chromosomes separates a little, but remain attached by chiasmata
- Diakinesis: chiasmata move to the ends of the chromosomes, nucleoli disappear, the nuclear membrane disintegrates, and the meiotic spindle begins to form

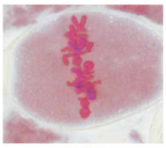
Cell Biology

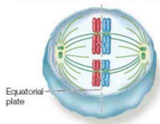
Cell Division: Cell division in Eukaryotes

Meiosis: Meiosis I

Metaphase I

- Spindle apparatus formed, and homologous chromosome pairs are attached to spindle fibres by kinetochores
- Homologous pairs of chromosomes arrange as a double row along the metaphase plate





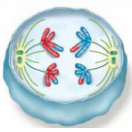
Cell Biology

Cell Division: Cell division in Eukaryotes

Meiosis: Meiosis I

Anaphase I

- The homologous chromosomes in each pair are separated and move to the opposite poles of the cell
- Chromosome arms are exchanged between the homologous chromosomes



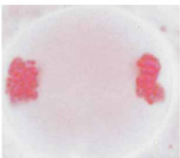
Cell Biology

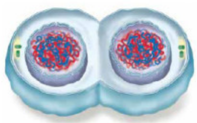
Cell Division: Cell division in Eukaryotes

Meiosis: Meiosis I

Telophase I

- The chromosomes become diffuse and the nuclear membrane reforms





Cell Biology

Cell Division: Cell division in Eukaryotes

Meiosis: Meiosis I

Cytokinesis

- Final cellular division to form two new cells, followed by Meiosis II
- Meiosis I is a reduction division

Cell Biology

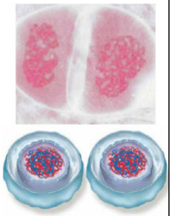
Cell Division: Cell division in Eukaryotes

Meiosis: Meiosis II

- Second meiotic division begins
- Meiosis II resembles normal mitotic division

Prophase II

- Nuclear membrane disintegrates, Nucleolus disappears, Chromosomes condense, mitotic spindle begins to form between centrioles



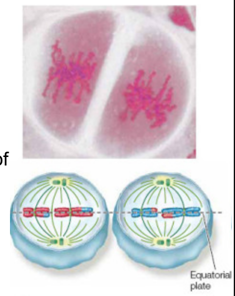
Cell Biology

Cell Division: Cell division in Eukaryotes

Meiosis: Meiosis II

Metaphase II

- Chromosomes are aligned at the metaphase plate
- Spindle fibers bind to both sides of centromeres



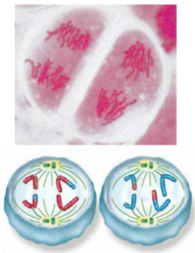
Cell Biology

Cell Division: Cell division in Eukaryotes

Meiosis: Meiosis II

Anaphase II

- Spindle fibers contract splitting the centromeres & moving the sister chromatids to opposite sides



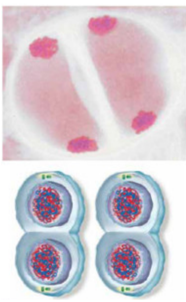
Cell Biology

Cell Division: Cell division in Eukaryotes

Meiosis: Meiosis II

Telophase II

- Finally, the nuclear envelope re-forms around the four sets of daughter chromosomes forming four daughter nuclei



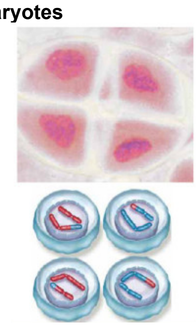
Cell Biology

Cell Division: Cell division in Eukaryotes

Meiosis: Meiosis II

Cytokinesis

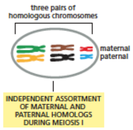
- Cells divide to make four daughter cells



Cell Biology

Cell Division: Cell division in Eukaryotes

Meiosis

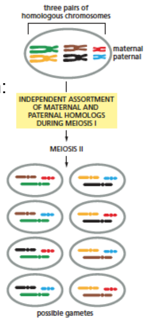


- Meiosis generates genetic diversity through:
  - the random alignment of maternal and paternal chromosomes in Meiosis I

Cell Biology

Cell Division: Cell division in Eukaryotes

Meiosis

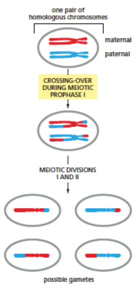


- Meiosis generates genetic diversity through:
  - the random alignment of maternal and paternal chromosomes in Meiosis I
  - the random alignment of the sister chromatids at Meiosis II

Cell Biology

Cell Division: Cell division in Eukaryotes

Meiosis



- Meiosis generates genetic diversity through:
  - the random alignment of maternal and paternal chromosomes in Meiosis I
  - the random alignment of the sister chromatids at Meiosis II
  - the exchange of genetic material between homologous chromosomes during Meiosis I

Cell Biology

Summary

- Cell division is the process in which a parent cell divides in to two or more cells
- Prokaryotic cell division involves binary fission
- Two types of cell division in eukaryotes: mitosis and meiosis
- Cell Cycle is the series of events that take place in a cell leading to duplication of its DNA and division to produce two daughter cells
- Cell cycle is regulated through three checkpoints that ensure proper division of the cell
- Checkpoints in the cell cycle are regulated by a complex of enzymes – Cyclin dependant kinases (Cdk)

Cell Biology

Summary

- **Mitosis** is a vegetative division where each daughter cell is genetically identical to the parent cell
- **Meiosis** is involved in producing **gametes** and results in four daughter cells with chromosomes half the number than the parent cells
- In meiosis the cell undergoes two divisions; meiosis I, meiosis II